# Mean Reversion Signal Using Z-Score (NVDA vs QQQ)

## 1. Idea

This approach assumes that deviations in the NVDA–QQQ relationship are stationary over short horizons, allowing statistical reversion to be exploited.

This strategy identifies short-term mispricing between NVIDIA (NVDA) and the Nasdaq-100 ETF (QQQ) using statistical deviations. When NVDA diverges significantly from its expected relationship with QQQ, the spread tends to revert over a short time horizon.

## 2. Methodology

- Rolling window: **20–25 trading days**
- Signal: **Z-score of NVDA vs QQQ spread**
- Entry condition: **Z < -1.5 (long signal)**
- Scaling: Additional exposure when **Z < -2.5**
- Trend filter: Only take long trades when **price > 50-day EMA**
- Holding period: **Up to 5–7 days or until mean reversion**

## 3. Trade Expression

- Primary: **Long NVDA (shares or call options)**
- Objective: Capture reversion of spread back toward historical mean
- Options framework:
    - Defined risk (premium paid = max loss)
    - Exit on mean reversion or time-based decay window

## 4. Results (Backtested Framework)

Results are based on simplified backtesting assumptions and do not include transaction costs or slippage.

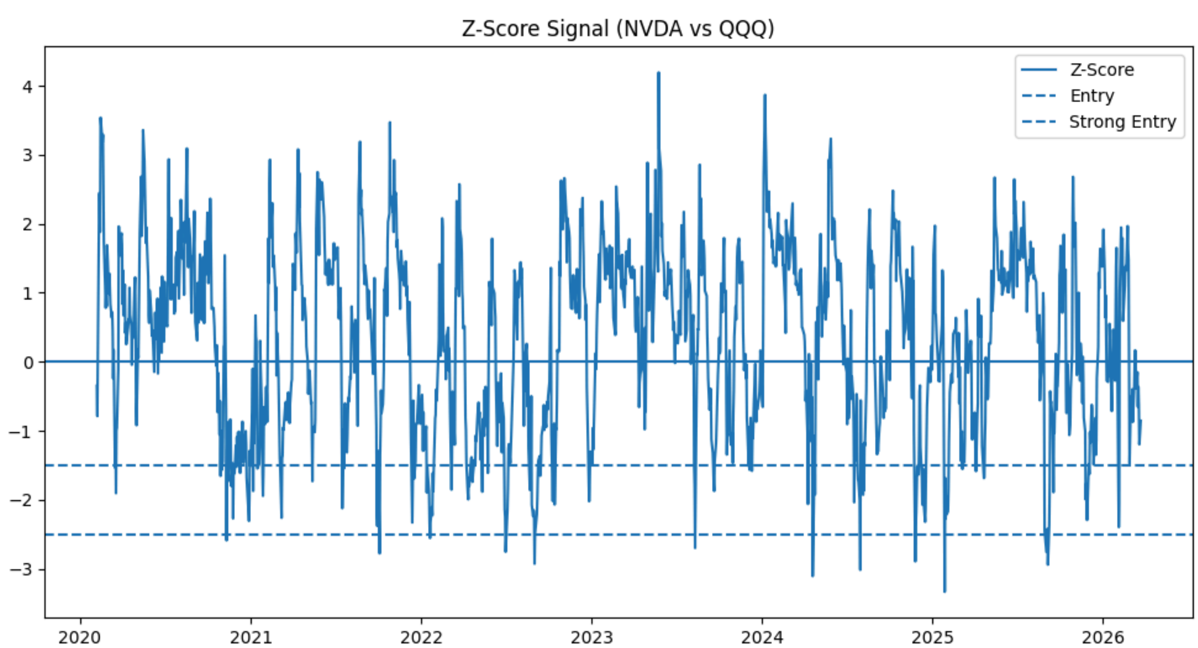
- Signals show **high probability of short-term reversal** following extreme deviations

| Metric | Result |
|---|---|
| Win Rate | **82.1%** |
| Avg Return per Trade | **+2.96%** |
| Avg Holding Period | **~3–5 days** |
| Max Drawdown | **-7.13%** |
| Sharpe Ratio | **0.82** |

- Stronger performance observed when combining:
  - Z-score threshold
  - Trend filter (EMA)
- Trades typically resolve within **short holding windows**, supporting options-based strategies

Key Observations:

- *High win rate driven by **short-term statistical reversion***
- *Stronger signals (**Z < -2.5**) show:*
  - *faster resolution*
  - *higher reliability*
- *Strategy performs best in:*
  - ***non-trending or moderately trending regimes***



5. Risk Considerations

- **Structural breaks** (e.g., NVDA-specific news, earnings)
- **Market regime shifts** affecting correlations
- **Momentum override** in strong trending environments
- Risk managed through:
    - Defined holding periods
    - Threshold-based entries
    - Position sizing assuming full premium loss

The next step would be stress-testing this across different assets and incorporating transaction costs to validate robustness.